IMAGE PROCESSING

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REFERENCE BOOK:

AN INTRODUCTION TO DIGITAL IMAGE

PROCESSING WITH MATLAB

ALASDAIR MCANDREW

INTRODUCTION

CHAPTER 1

LECTURE 1 AND 2

1.1 IMAGES AND PICTURES

- Human beings are predominantly visual creatures
- For our purposes, an image is a single picture which represents something. It may be a picture of a person, of people or animals, or of an outdoor scene, or a microphotograph of an electronic component, or the result of medical imaging. Even if the picture is not immediately recognizable, it will not be just a random blur.

CLASSICAL IMAGE PROCESSING

- Image processing is a subset of computer vision. A computer vision system uses the image processing algorithms to try and perform emulation of vision at human scale. For example, if the goal is to enhance the image for later use, then this may be called image processing. And if the goal is to recognise objects, defect for automatic driving, then it can be called computer vision.

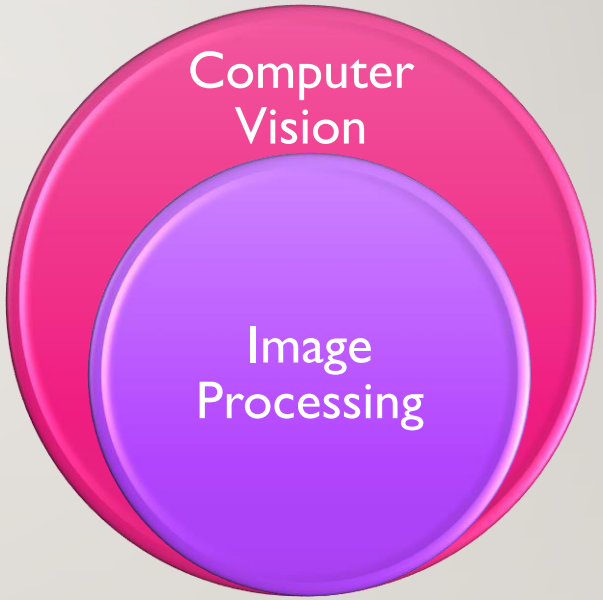


IMAGE PROCESSING EXAMPLES

- Image Denoising
- Image Restoration



(a) The original image (b) After removing the blur

Figure 1.3: Image deblurring



(a) The original image (b) After removing noise

Figure 1.2: Removing noise from an image

IMAGE PROCESSING AND COMPUTER VISION

Image



Image Processing

Image in different Format



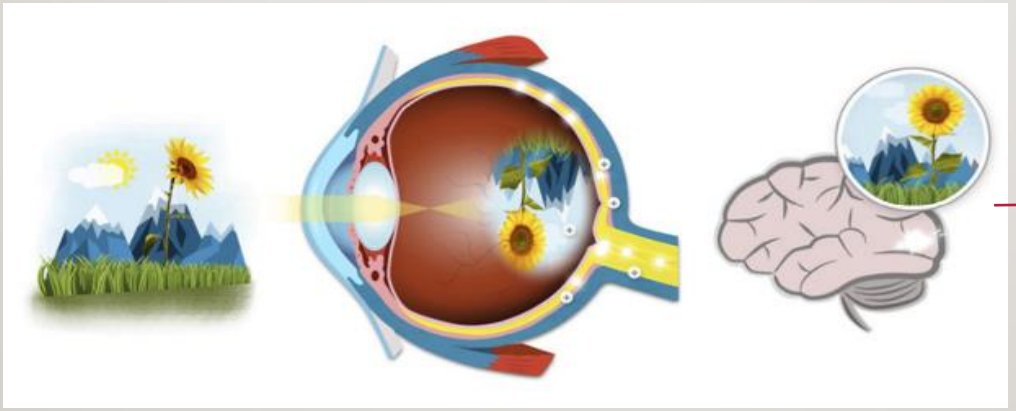
Image



Computer Vision

Knowledge of the scene **"Smiling Kid"**

WHAT IS THE DIFFERENCE BETWEEN HUMAN VISION AND COMPUTER VISION



Mountain
Sun
Cloud
Grass
flower



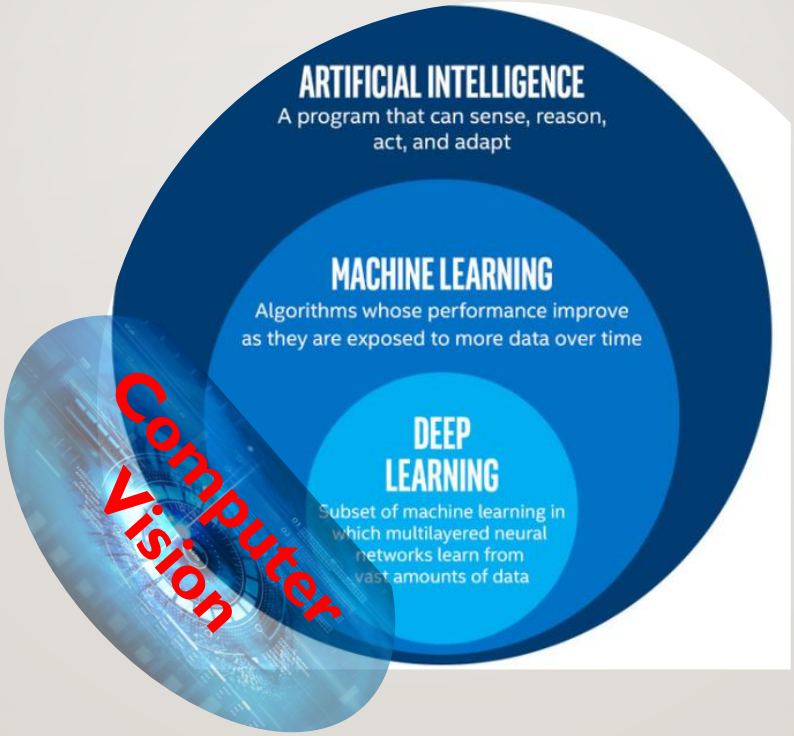


Sensors

Inference System

Mountain
Sun
Cloud
Grass
flower

COMPUTER VISION AND ARTIFICIAL INTELLIGENCE








COMPUTER VISION IN LIFE : CROWD DETECTION AND ABNORMAL BEHAVIOR DETECTION

- CCTV cameras are everywhere around us in offices, roads, hospitals, banks, train stations, parking lots, etc and 24/7 surveillance is difficult. The computer vision techniques allow us to monitor the events in real-time and detect anomalies or any specific action detection. **Automatic Number Plate Recognition (ANPR)** system can be used to control automatic gates, vehicle tracking, analyzing crowd and counting number of people.

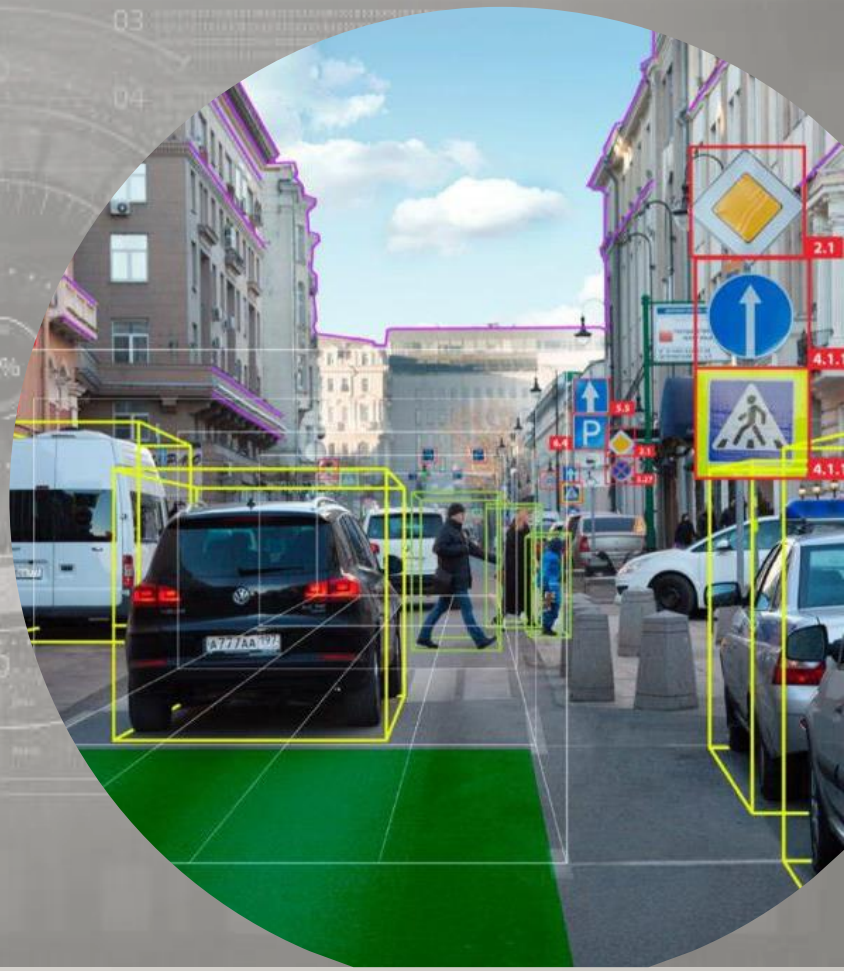


REAL-TIME ABNORMAL CROWD BEHAVIOUR DETECTION



COMPUTER VISION IN LIFE : EVENY DETECTION

Artificial Intelligence is on the boom right now and companies are investing in self-driving technology. Computer vision is used to detect lanes and find a path for autonomous vehicles. Information from various sensors is analyzed and used to detect objects on the path like traffic lights, traffic signs and according to obstacles, we decide the appropriate action that needs to be taken.



TESLA'S DRIVERLESS SYSTEM DETECTING OBJECTS IN A FOGGY SCENARIO VIA SOURCE



COMPUTER VISION IN LIFE : MEDICAL IMAGE PROCESSING

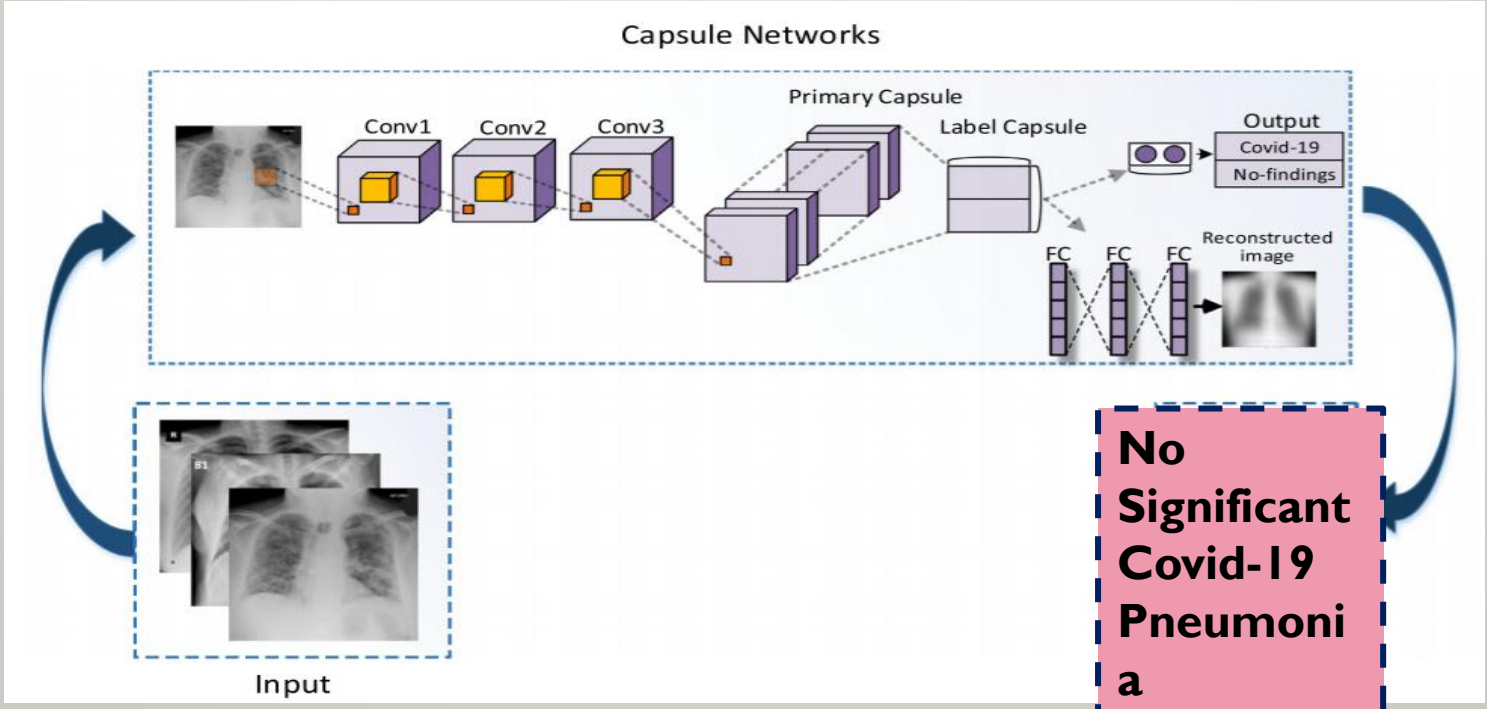




The progress in computer vision has led to extensive use of the medical imaging data to provide us with better prediction, diagnosis, and treatment of diseases. Some **examples** of this are the detection of tumours, arteriosclerosis or other malign changes, measurement of organ dimensions, blood flow, enhancement of ultrasonic or X-ray images that are interpreted by humans.

COMPUTER VISION IN LIFE : MEDICAL IMAGE PROCESSING

DETECTING COVID-19









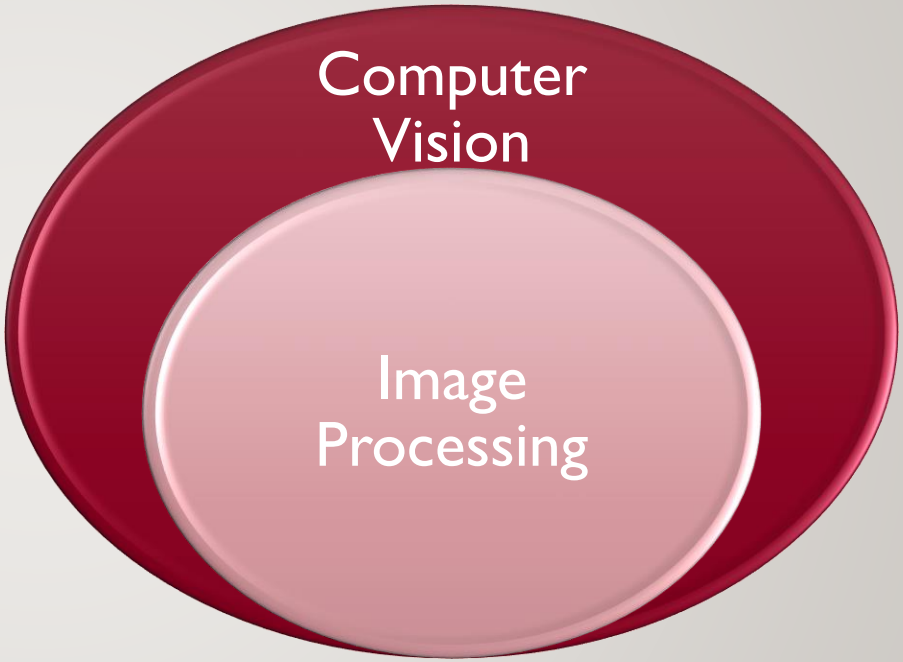


COMPUTER VISION IN LIFE : AUTOMATIC INSPECTION OF OBJECTS

• Automatic inspection of objects and classifying them into different categories play a major role in manufacturing industries. Industrial robots use computer vision algorithms to perform various tasks like separating two or more different objects into their respective categories, detecting whether the product is labelled or not. If not, then we can reject the product on the conveyor belt.

CLASSICAL IMAGE PROCESSING

- Image processing is a subset of computer vision. A computer vision system uses the image processing algorithms to try and perform emulation of vision at human scale. For example, if the goal is to enhance the image for later use, then this may be called image processing. And if the goal is to recognise objects, defect for automatic driving, then it can be called computer vision.

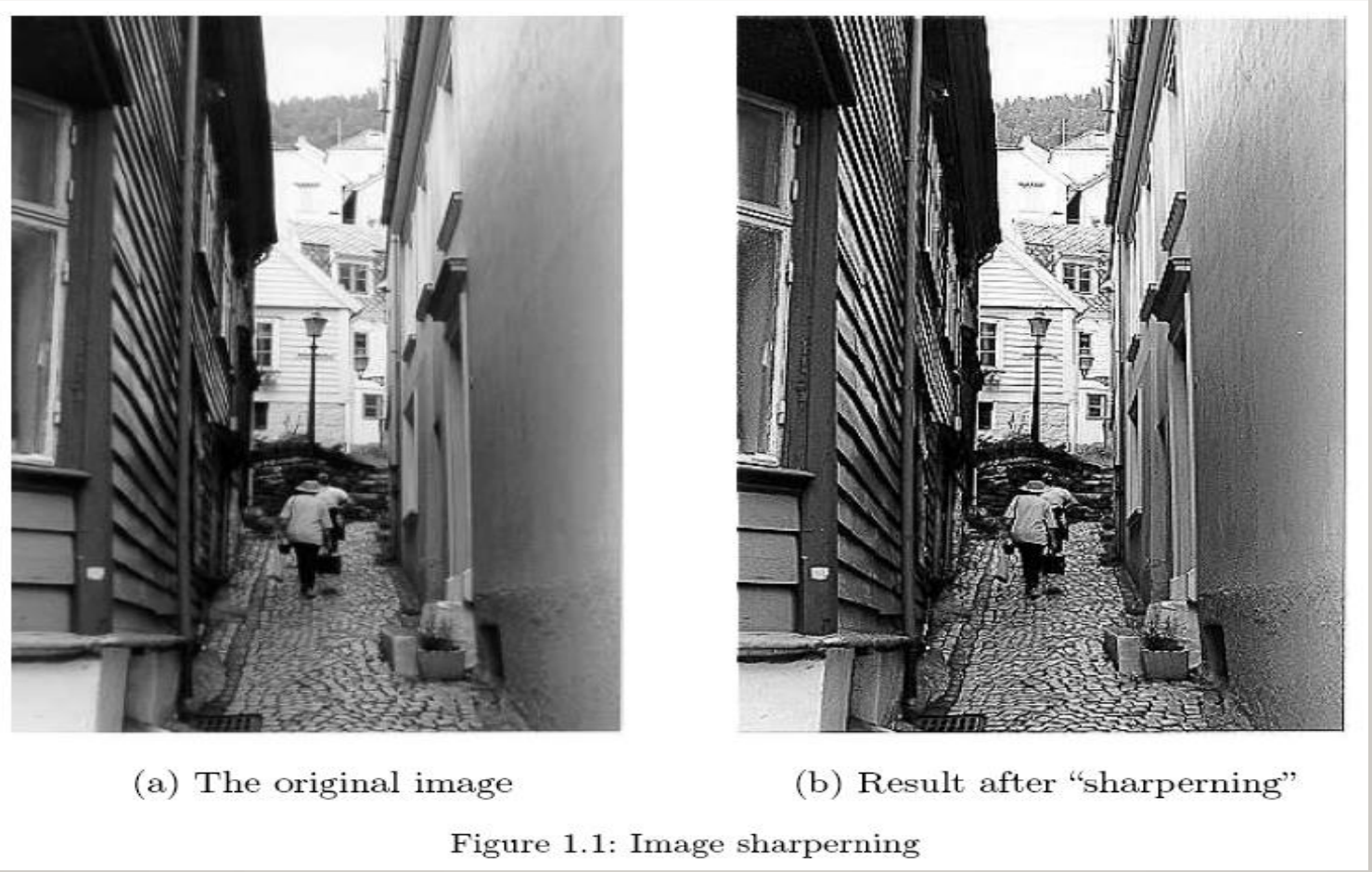


1.2 WHAT IS IMAGE PROCESSING?

Image processing involves changing the nature of an image in order to either

1. improve its pictorial information for human interpretation.
2. render it more suitable for autonomous machine perception.

•ENHANCING THE EDGES OF AN IMAGE TO MAKE IT APPEAR SHARPER



(a) The original image

(b) Result after "sharperning"

Figure 1.1: Image sharperning

•REMOVING NOISE FROM AN IMAGE



(a) The original image



(b) After removing noise

Figure 1.2: Removing noise from an image

•REMOVING MOTION BLUR FROM AN IMAGE



(a) The original image

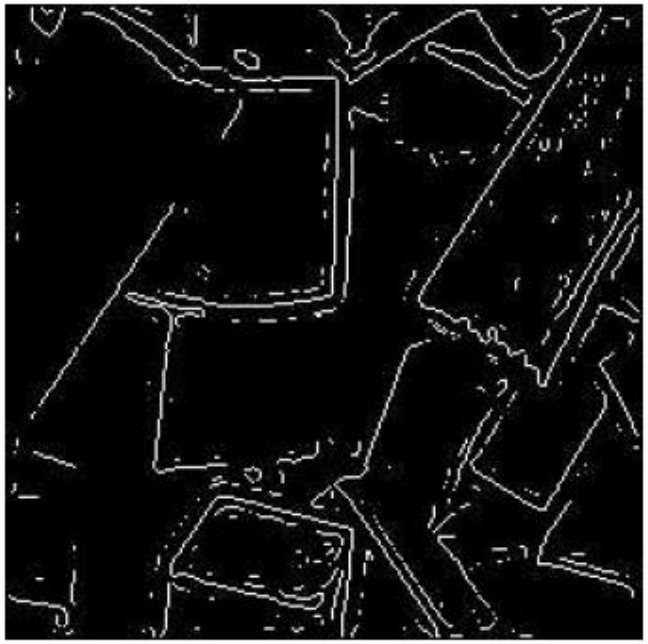
(b) After removing the blur

Figure 1.3: Image deblurring

•OBTAINING THE EDGES OF AN IMAGE



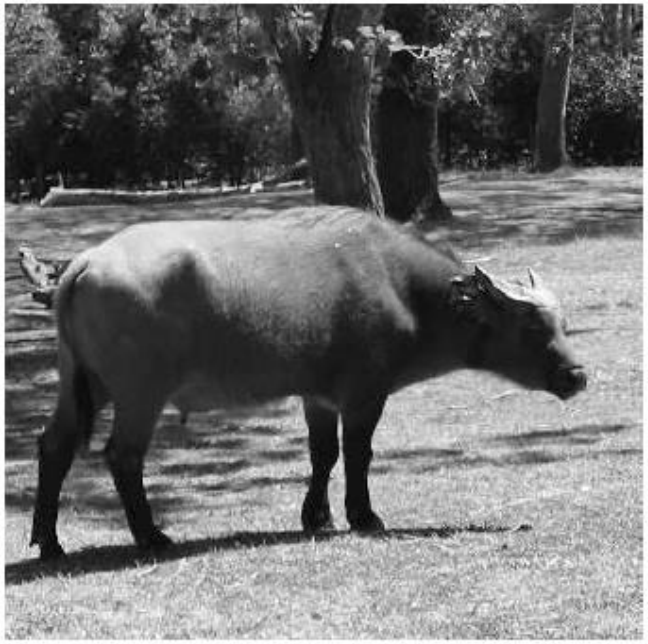
(a) The original image



(b) Its edge image

Figure 1.4: Finding edges in an image

•REMOVING DETAIL FROM AN IMAGE



(a) The original image



(b) Blurring to remove detail

Figure 1.5: Blurring an image

IMAGE PROCESSING AND COMPUTER VISION

Image



Image Processing

Image in different Format



Image



Computer Vision

Knowledge of the scene

"Smiling Kid"

IMAGE ACQUISITION AND SAMPLING

- Sampling refers to the process of digitizing a continuous function. For example, suppose we take the function

$$y = \sin(x) + \frac{1}{3}\sin(3x).$$

Nyquist criterion - sampling theorem
The sampling frequency is at least twice the maximum frequency I
n the function

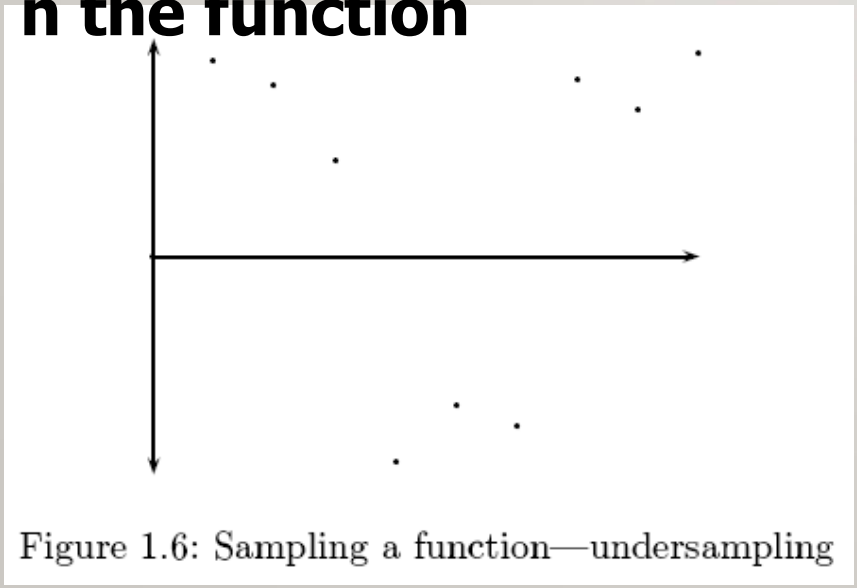


Figure 1.6: Sampling a function—undersampling

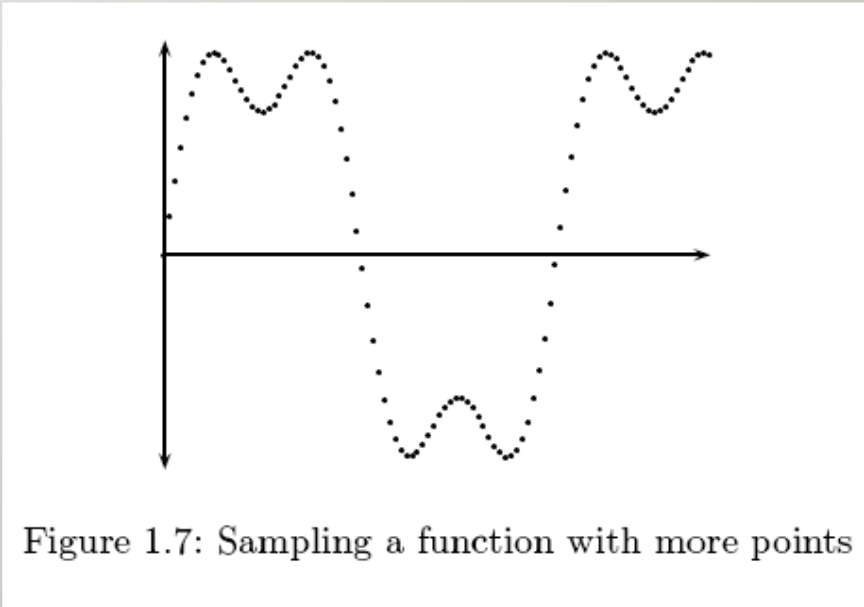
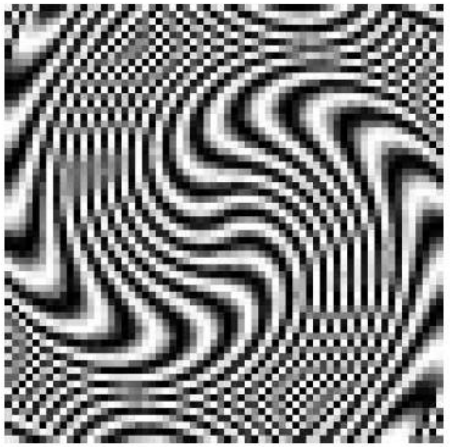


Figure 1.7: Sampling a function with more points

EXAMPLE OF AN IMAGE AND ITS UNDERSAMPLING VERSION



Correct sampling; no aliasing



An undersampled version with aliasing

Figure 1.8: Effects of sampling

USING LIGHT

- Light is the predominant energy source for images; simply because it is the energy source which human beings can observe directly. We are all familiar with photographs, which are a pictorial record of a visual scene.
- Many digital images are captured using visible light as the energy source; this has the advantage of being safe, cheap, easily detected and readily processed with suitable hardware.

TWO VERY POPULAR METHODS OF PRODUCING A DIGITAL IMAGE ARE :

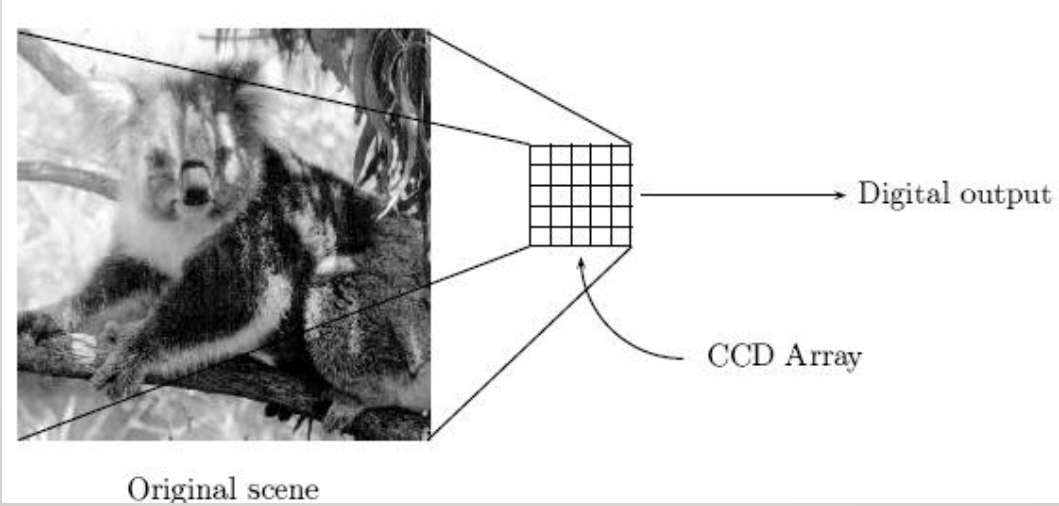
Digital camera

- CCD ”Charge Coupled Device“ camera
- CMOS ”Complementary mتtal oxide semi-conductor ”camera

Flat-bed scanner.

CCD CAMERA

CCD ”Charge Coupled Device“ it consists of an array of light snsetive cell called photosites each of which produce a voltage proportional to the intensity of light falling on them.
It produces images with high resolution and robust against noise



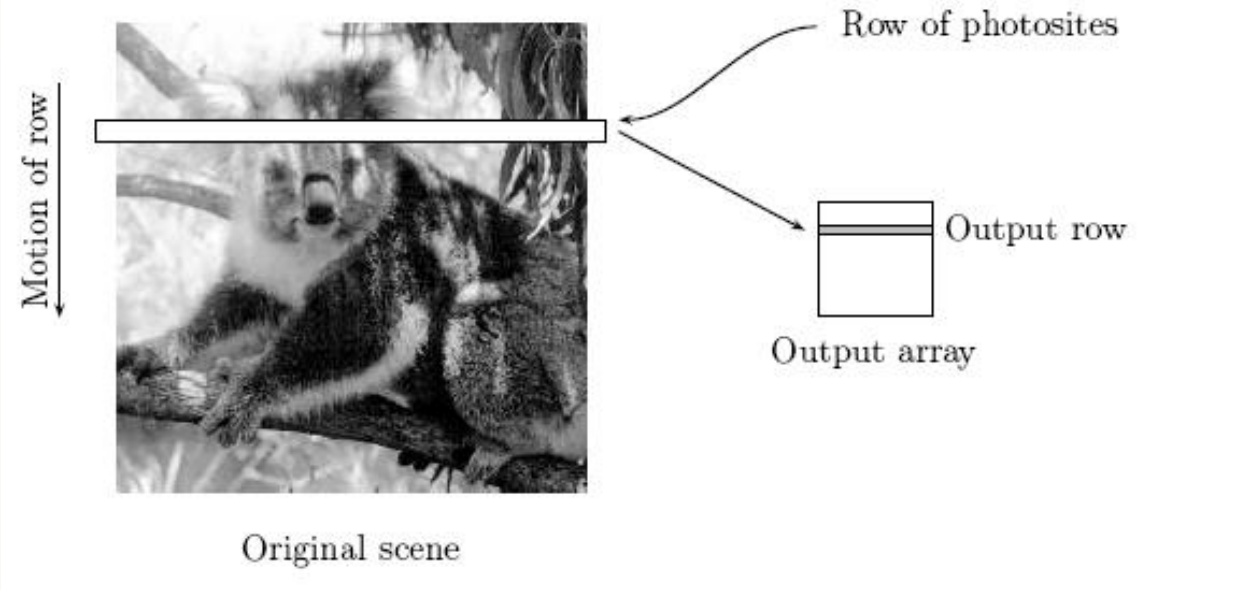
Capturing an image with a CCD array

CMOS CAMERA

- Cheaper product require less power to run than CCD but more susceptible to noise.

FLAT BED SCANNER.

This works on a principle similar to the CCD camera. Instead of the entire image being captured at once on a large array, a single row of photosites is moved across the image, capturing it row-by-row as it moves



Capturing an image with a CCD scanner

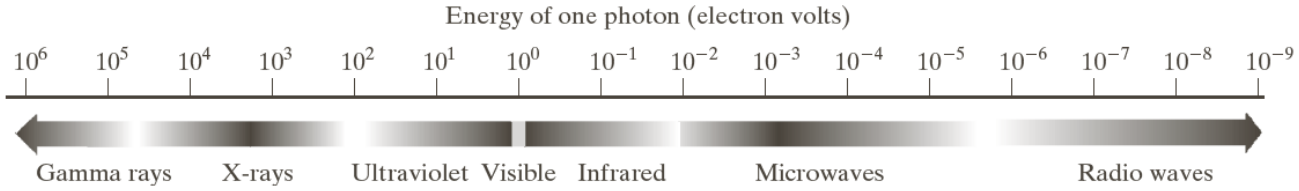
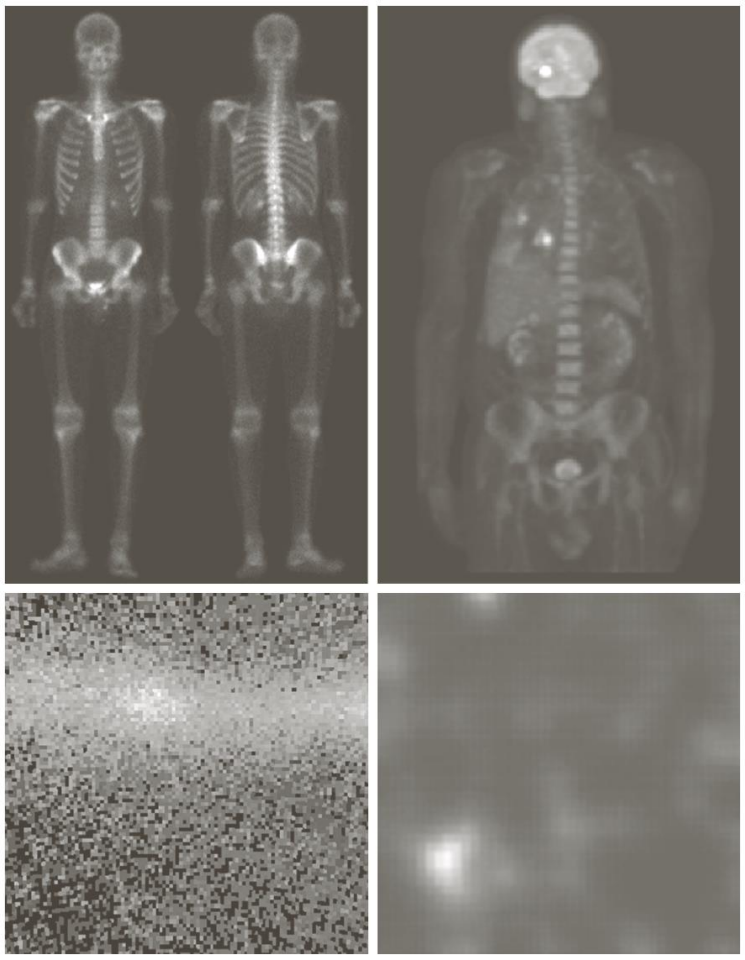
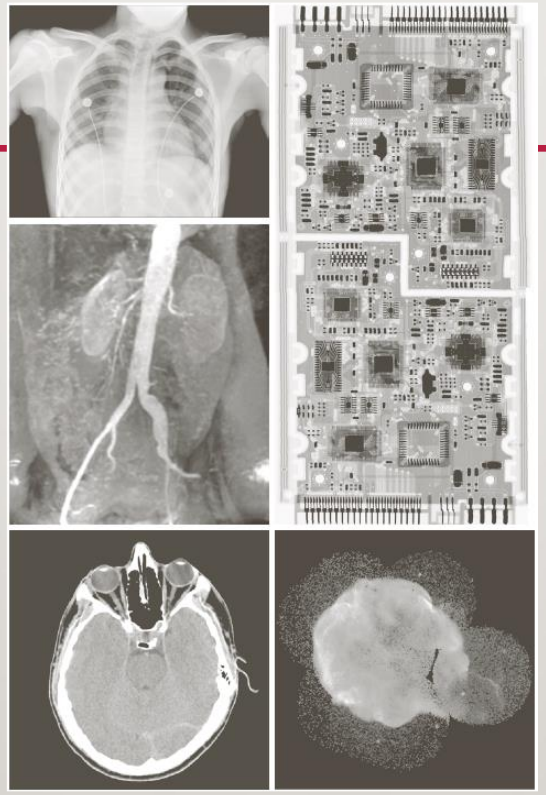


FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.



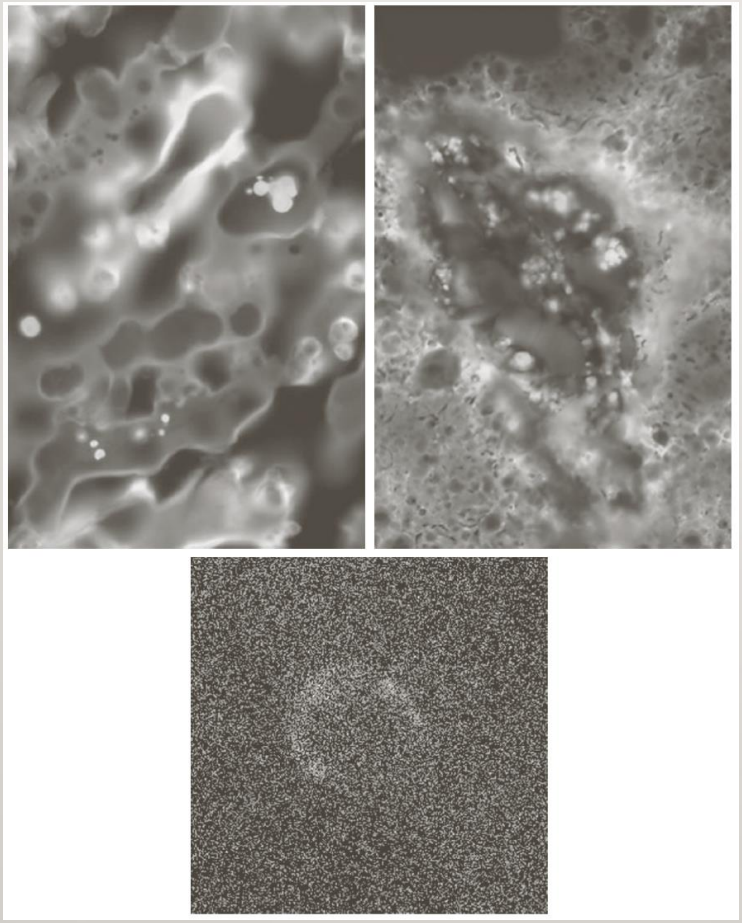
a b
c d

FIGURE 1.6
Examples of gamma-ray imaging. (a) Bone scan. (b) PET image. (c) Cygnus Loop. (d) Gamma radiation (bright spot) from a reactor valve. (Images courtesy of (a) G.E. Medical Systems, (b) Dr. Michael E. Casey, CTI PET Systems, (c) NASA, (d) Professors Zhong He and David K. Wehe, University of Michigan.)



a b c d e

FIGURE 1.7 Examples of X-ray imaging. (a) Chest X-ray. (b) Aortic angiogram. (c) Head CT. (d) Circuit boards. (e) Cygnus Loop. (Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center; (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School; (d) Mr. Joseph E. Pascente, Lixi, Inc.; and (e) NASA.)



a b
c

FIGURE 1.8
Examples of ultraviolet imaging.
(a) Normal corn.
(b) Smut corn.
(c) Cygnus Loop.
(Images courtesy of (a) and (b) Dr. Michael W. Davidson, Florida State University, (c) NASA.)

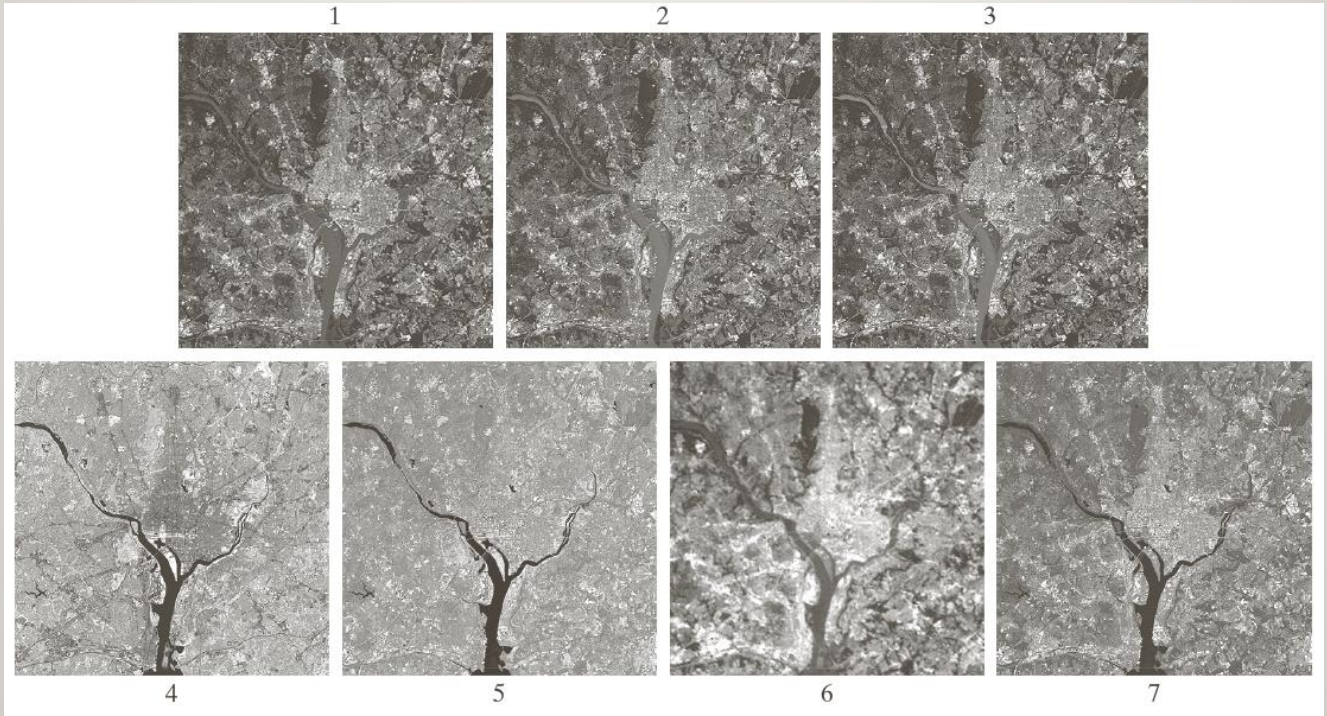


FIGURE 1.10 LANDSAT satellite images of the Washington, D.C. area. The numbers refer to the thematic bands in Table 1.1. (Images courtesy of NASA.)

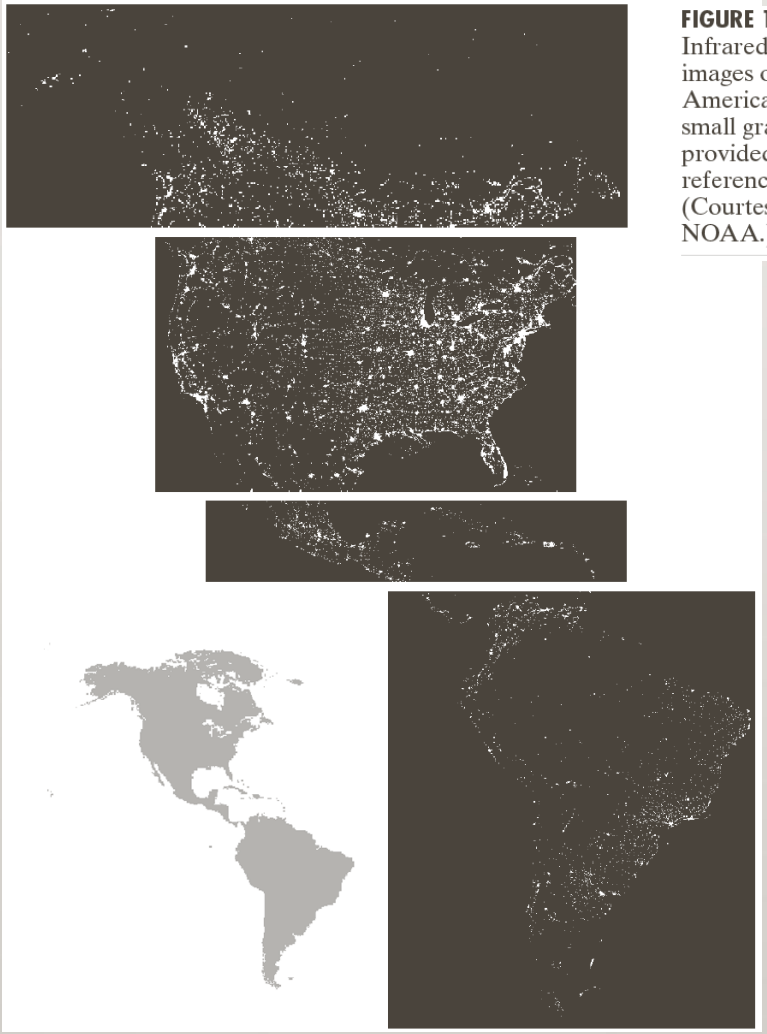
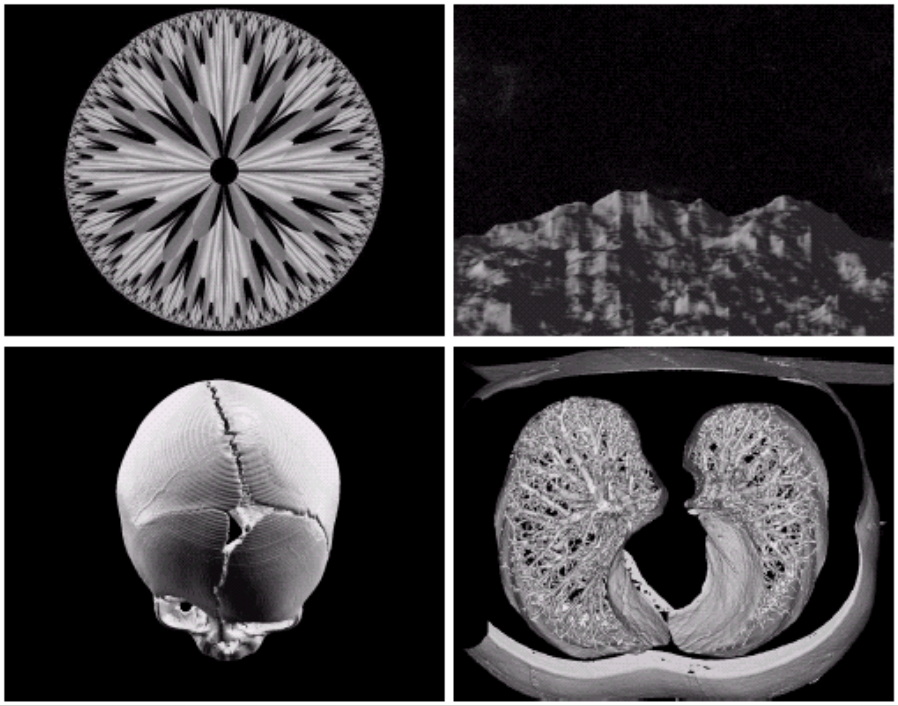


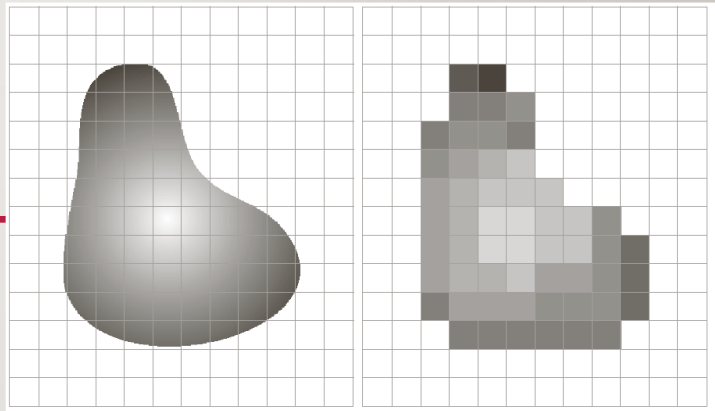
FIGURE 1.12
Infrared satellite images of the Americas. The small gray map is provided for reference. (Courtesy of NOAA.)



a b
c d

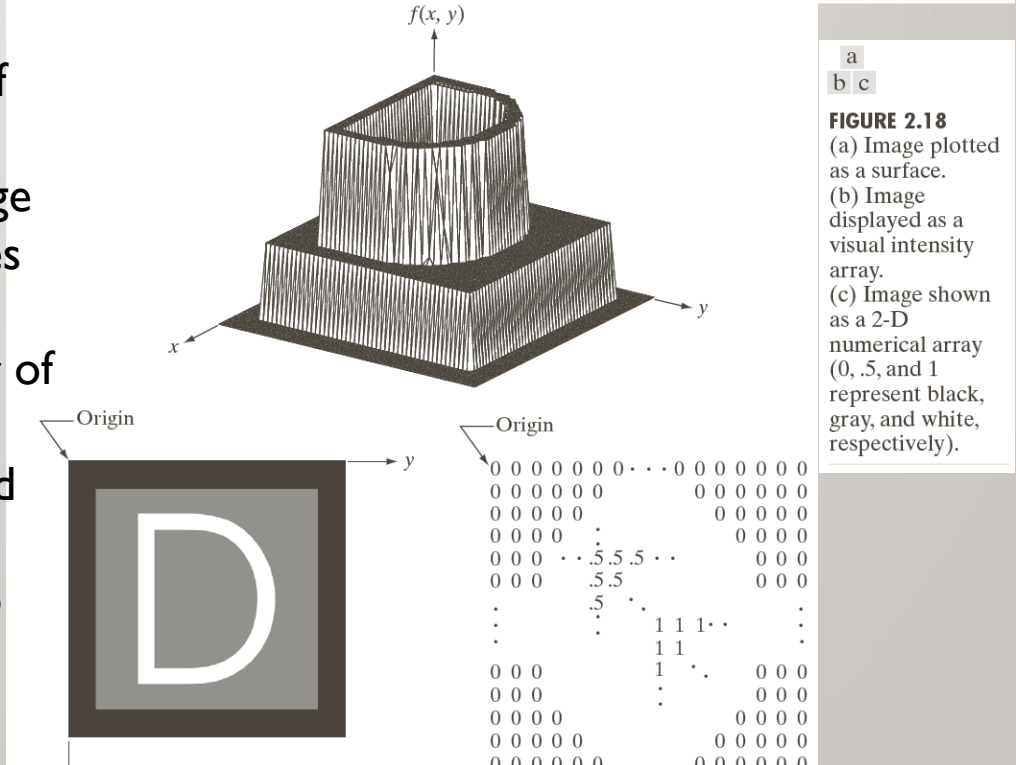
FIGURE 1.22
(a) and (b) Fractal images. (c) and (d) Images generated from 3-D computer models of the objects shown. (Figures (a) and (b) courtesy of Ms. Melissa D. Binde, Swarthmore College, (c) and (d) courtesy of NASA.)

An image may be defined as a two-dimensional function, f (x , y), where x and y are spatial (plane) coordinates, and the amplitude of ***f*** at any pair of coordinates (x , y) is called the intensity or gray level of the image at that point. When x, y, and the intensity values of f are all finite, discrete quantities, we call the image a digital image. The field of digital image processing refers to processing digital images by means of a digital computer. Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels. Pixel is the term used most widely to denote the elements of a digital image



a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.



a
b c

FIGURE 2.18
(a) Image plotted as a surface.
(b) Image displayed as a visual intensity array.
(c) Image shown as a 2-D numerical array (0, .5, and 1 represent black, gray, and white, respectively).

STORING IMAGES

Images are stored in computers using two different techniques: raster graphics and vector graphics.

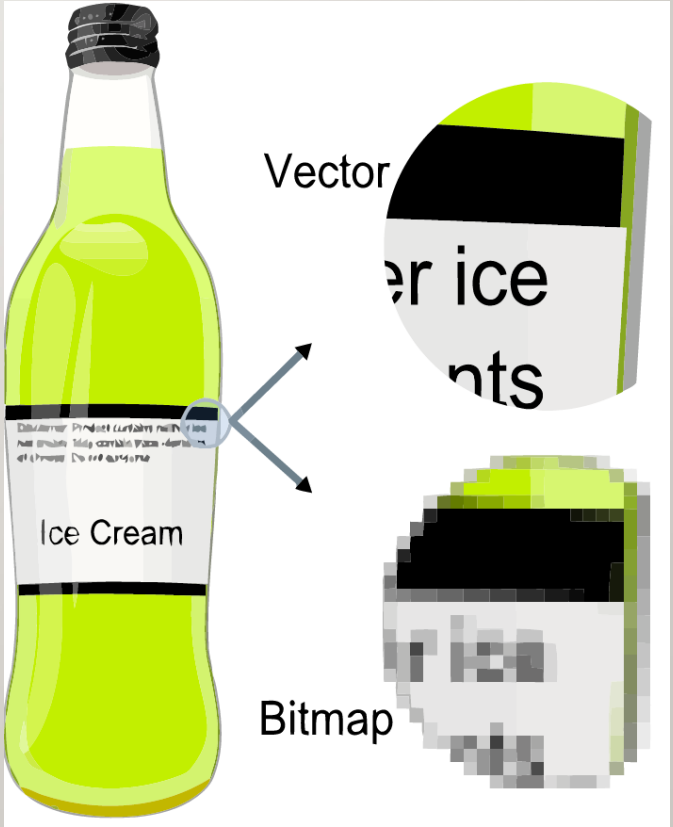
Raster graphics

Raster graphics (or **bitmap graphics**) are used when we need to store an analog image such as a photograph. A photograph consists of analog data, similar to audio information. The difference is that the intensity (color) of data varies in space instead of in time. This means that data must be sampled. However, sampling in this case is normally called **scanning**. The samples are called **pixels** (picture elements).

REPRESENTING IMAGE

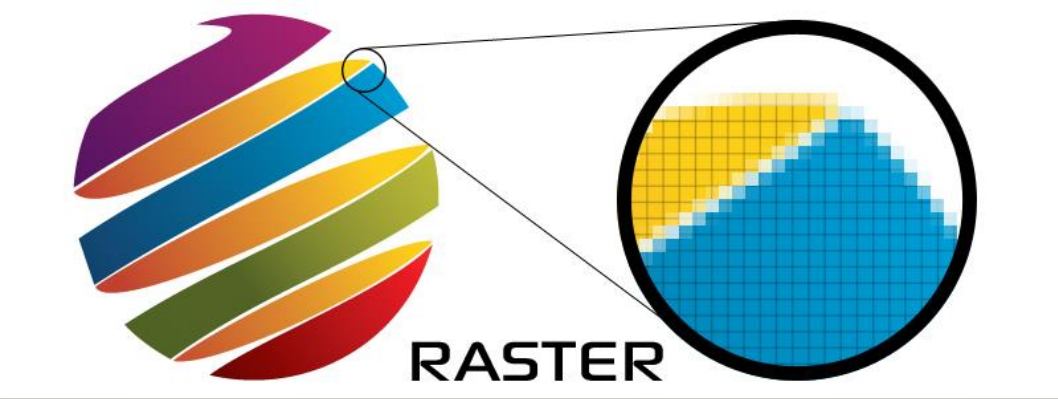
- Images are stored in computers using two different techniques:
  - Raster/Bitmap graphics
  - vector graphics

  Check out this video



REPRESENTING IMAGE

- Raster graphics are bitmaps. A bitmap is a grid of individual pixels (picture element) that collectively compose an image.
  - To maximize the quality of a raster image, you must keep in mind that the raster format is resolution-specific. Resolution is measured in dpi, or dots per inch. The higher the dpi, the better the resolution.



IMAGES AND DIGITAL IMAGES

We may consider this image as being a two dimensional function, where the function values give the brightness of the image at any given point.

A digital image difers from a photo in that the x ,y , and f(x,y) values are all discrete. Usually they take on only integer values, so the image shown in figure 1.13 will have x and y ranging from 1 to 256 each, and the brightness values also ranging from 0 (black) to 255 (white).

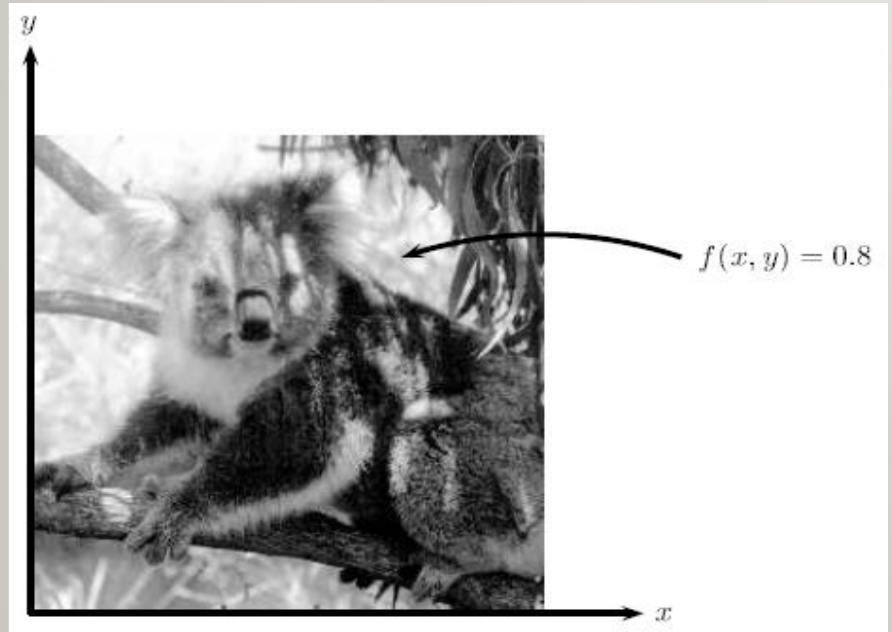


Figure 1.13: An image as a function

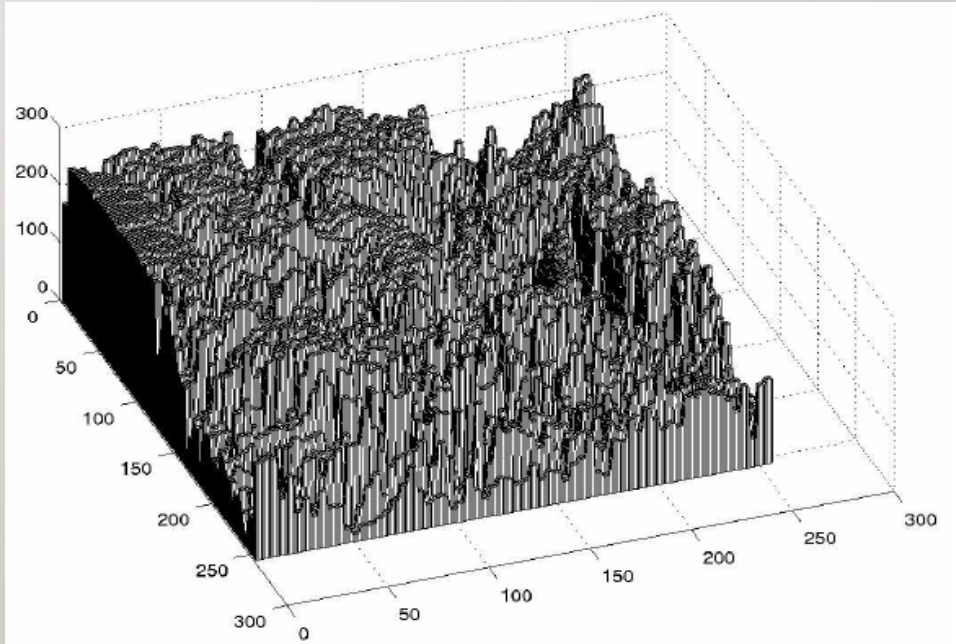


Figure 1.14: The image of figure 1.13 plotted as a function of two variables

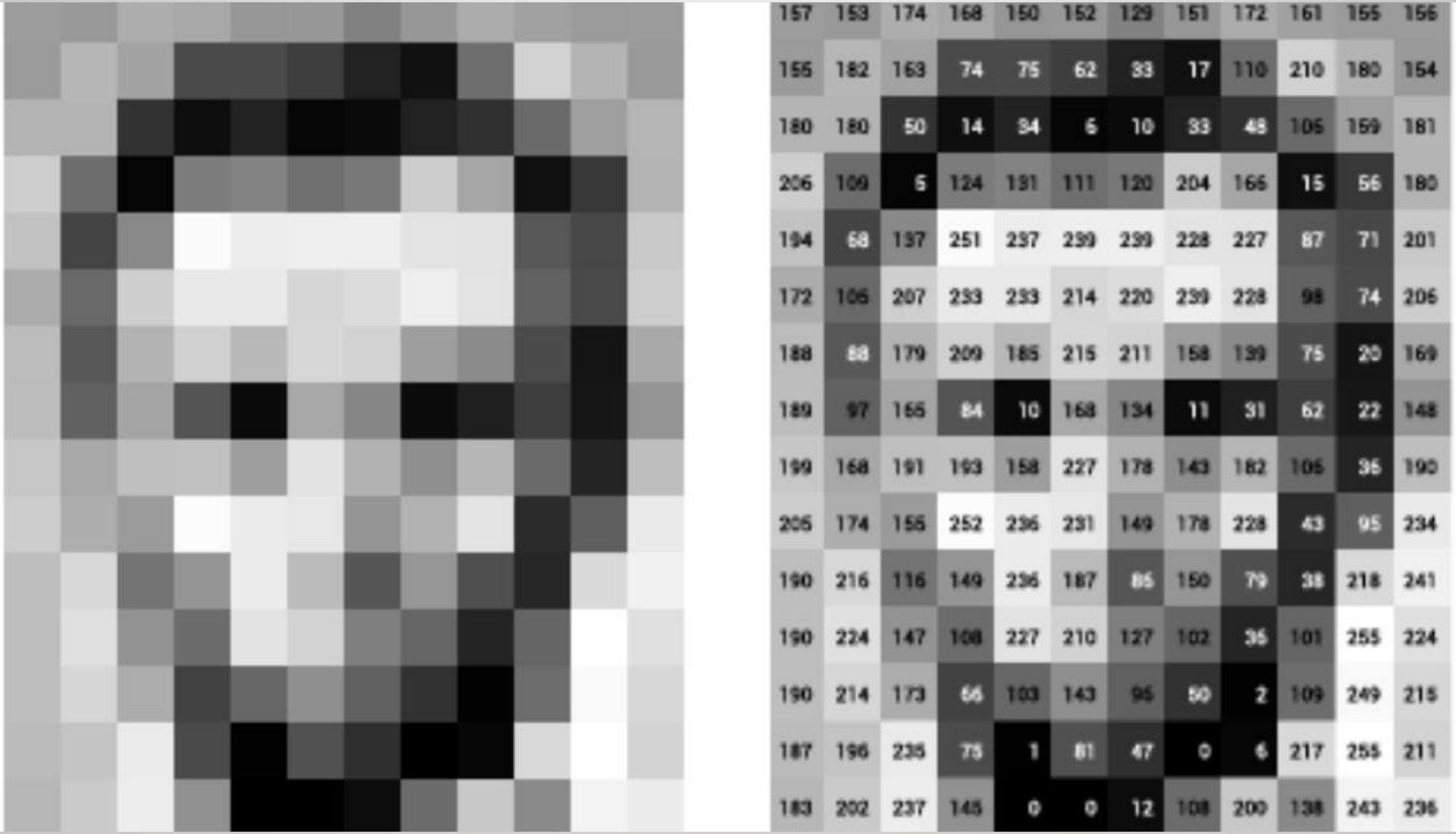


IMAGE REPRESENTATION

44 REPRESENTING IMAGE

- Sometimes it is very useful to be able to change the size of an image without losing quality. This **font** for example can be made in many different sizes.

small **bigger** **Bigger**

- Notice that the shape of each letter is still smooth no matter what the size. You can make it as large as you like and it will not pixelate or go fuzzy as a bitmap image would. This is an example of a Vector graphics →

PIXELS, WITH A NEIGHBOURHOOD

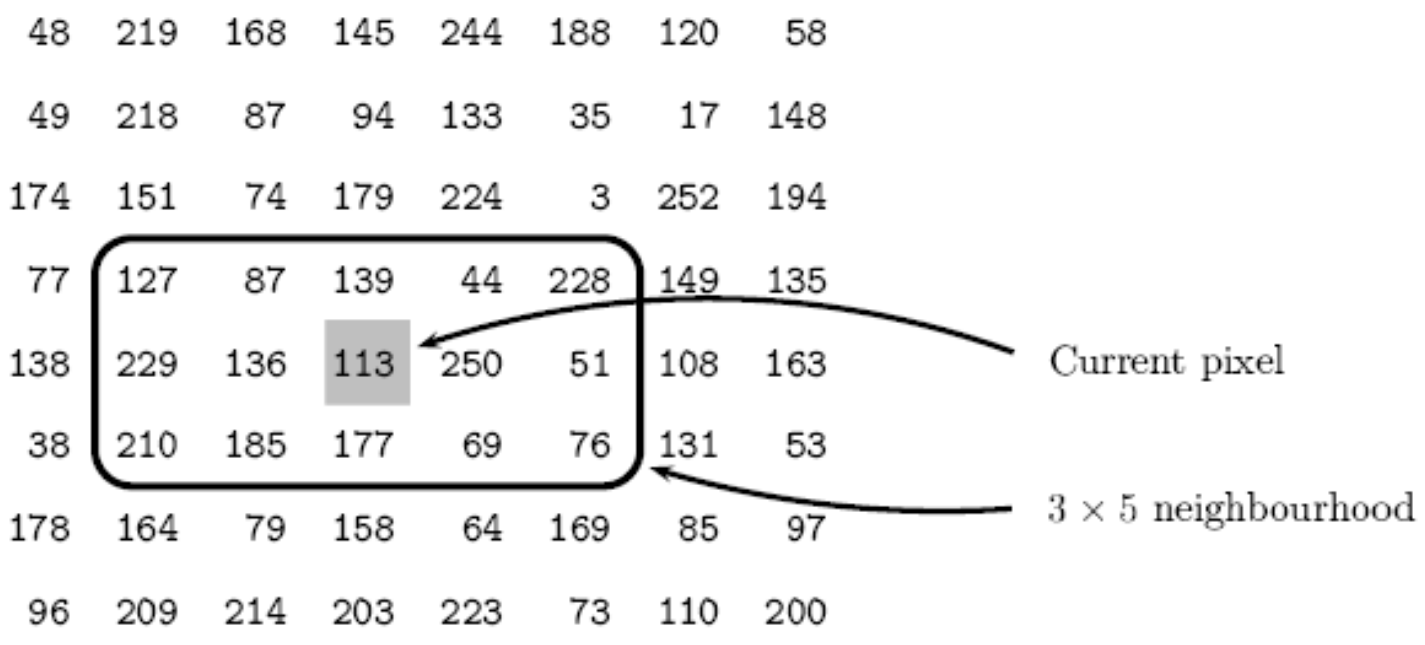


Figure 1.15: Pixels, with a neighbourhood

REPRESENTING IMAGE

- Vector graphics
  - This method does not store the bit patterns for each pixel. Vector graphics are based on mathematical formulas that define geometric primitives such as polygons, lines, curves, circles and rectangles.



47 REPRESENTING IMAGE

- Vector graphics
  - Instead of storing every single dot in the picture as a bitmap, vector images store a set of instructions of HOW to draw the shape. For example, consider the line below - this is a vector graphic:

a _____ **b**

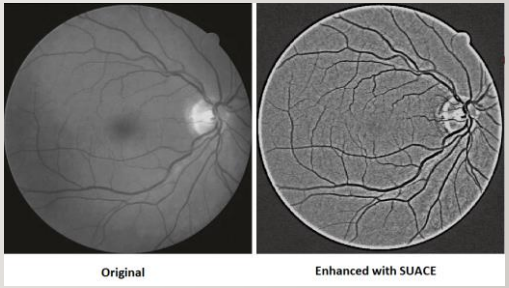
1. draw a dot at point "a"
2. draw a dot at point "b"
3. connect the points together with a straight line of a certain thickness
4. color the line red

The size of the file does not change if the image needs to be larger - the same instructions will make the image as large as you like, and yet the line would always remain smooth and clear.

ASPECTS OF IMAGE PROCESSING

- **Image enhancement.** This refers to processing an image so that the result is more suitable for a particular application. Example include:
  - sharpening or de-blurring an out of focus image,
  - highlighting edges,
  - improving image contrast, or brightening an image,
  - Remove noise



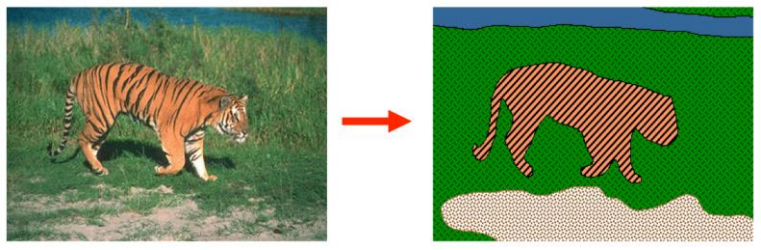


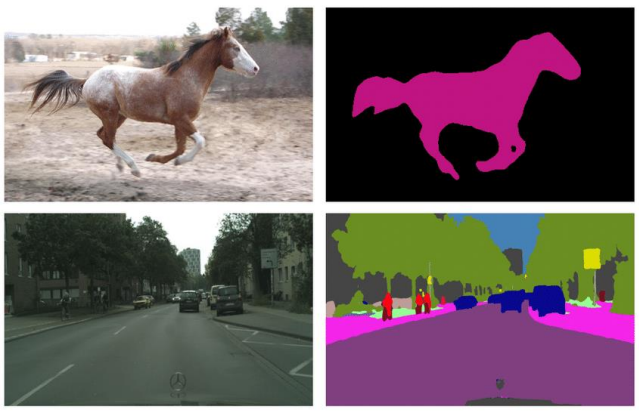
ASPECTS OF IMAGE PROCESSING

- **Image restoration.** This may be considered as reversing the damage done to an image by a known cause, for example:
  - removing of blur caused by linear motion,
  - removal of optical distortions,
  - removing periodic interference.



ASPECTS OF IMAGE PROCESSING





- **Image segmentation.** This involves subdividing an image into constituent parts, or isolating certain aspects of an image:
  - finding lines, circles, or particular shapes in an image,
  - in an aerial photograph, identifying cars, trees, buildings, or roads (semantic segmentation).

1.7 AN IMAGE PROCESSING TASK

- Acquiring the image.
- Preprocessing
- Segmentation.
- Representation and description.

HOW DOES A COMPUTER SEE?





Sensors

Inference System

Mountain
Sun
Cloud
Grass
flower





Sensors

Pre-processing

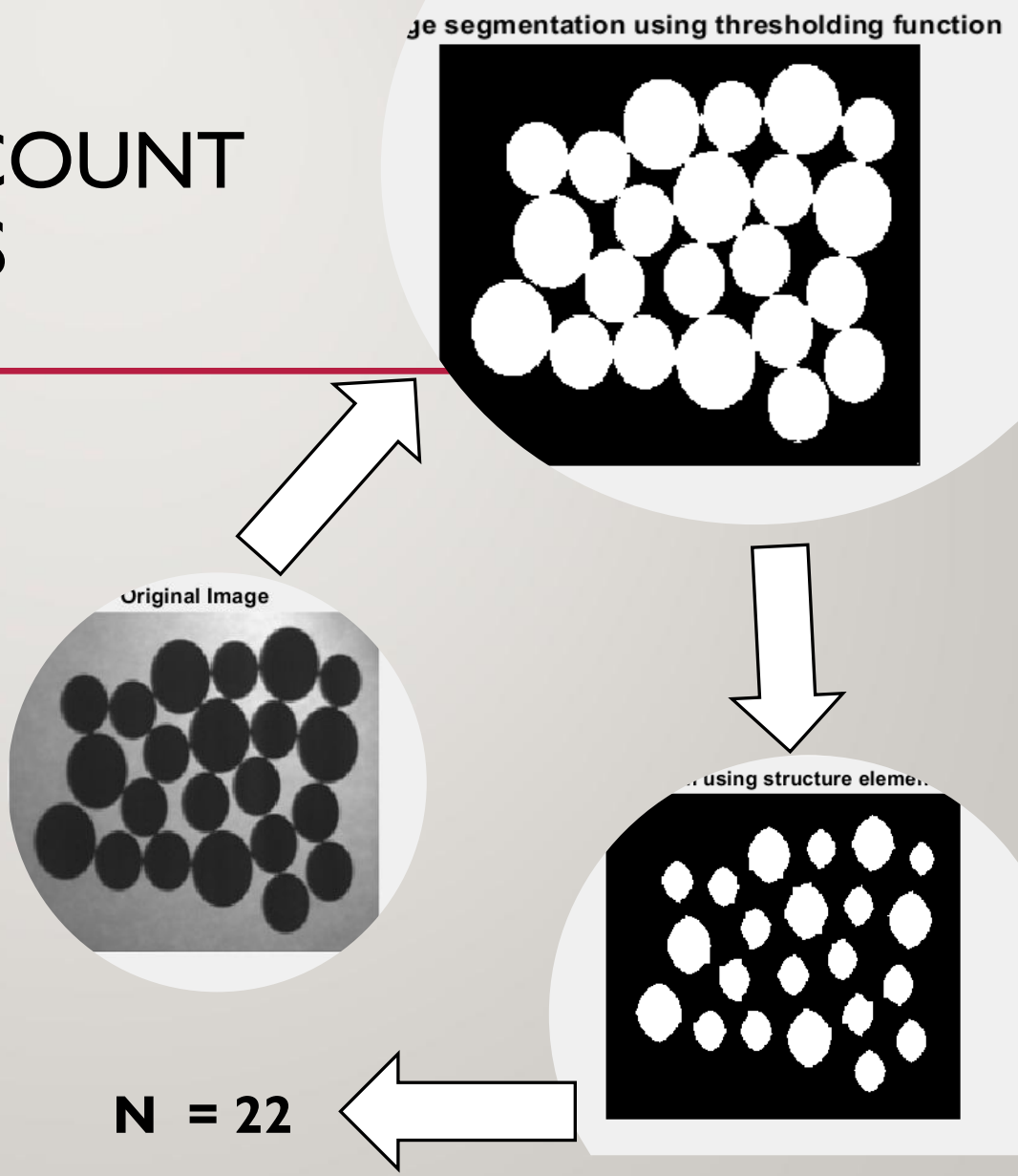
Feature Extraction

Machine Learning Model

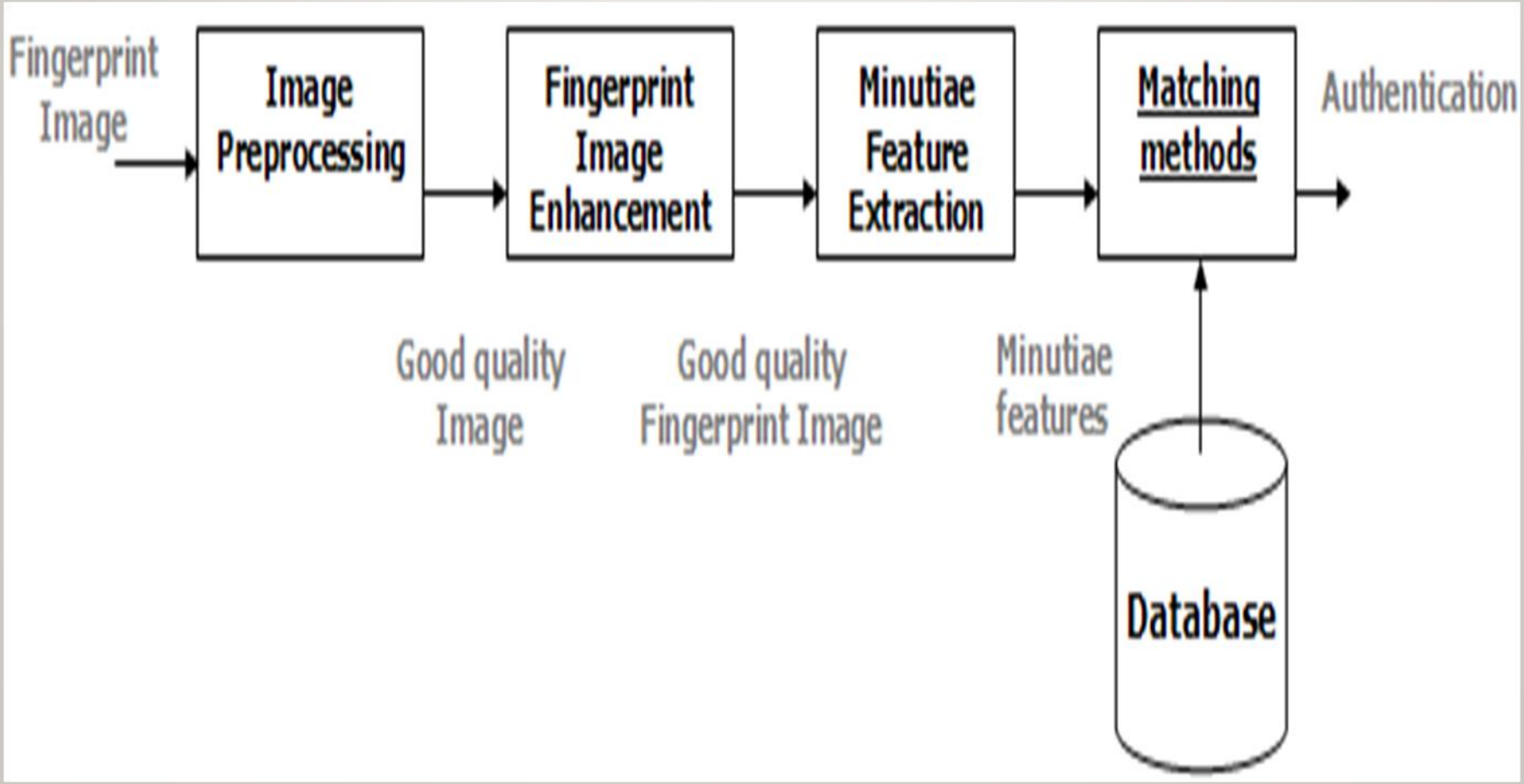
Mountain
Sun
Cloud
Grass
flower

SIMPLE EXAMPLE : COUNT NUMBER OF COINS

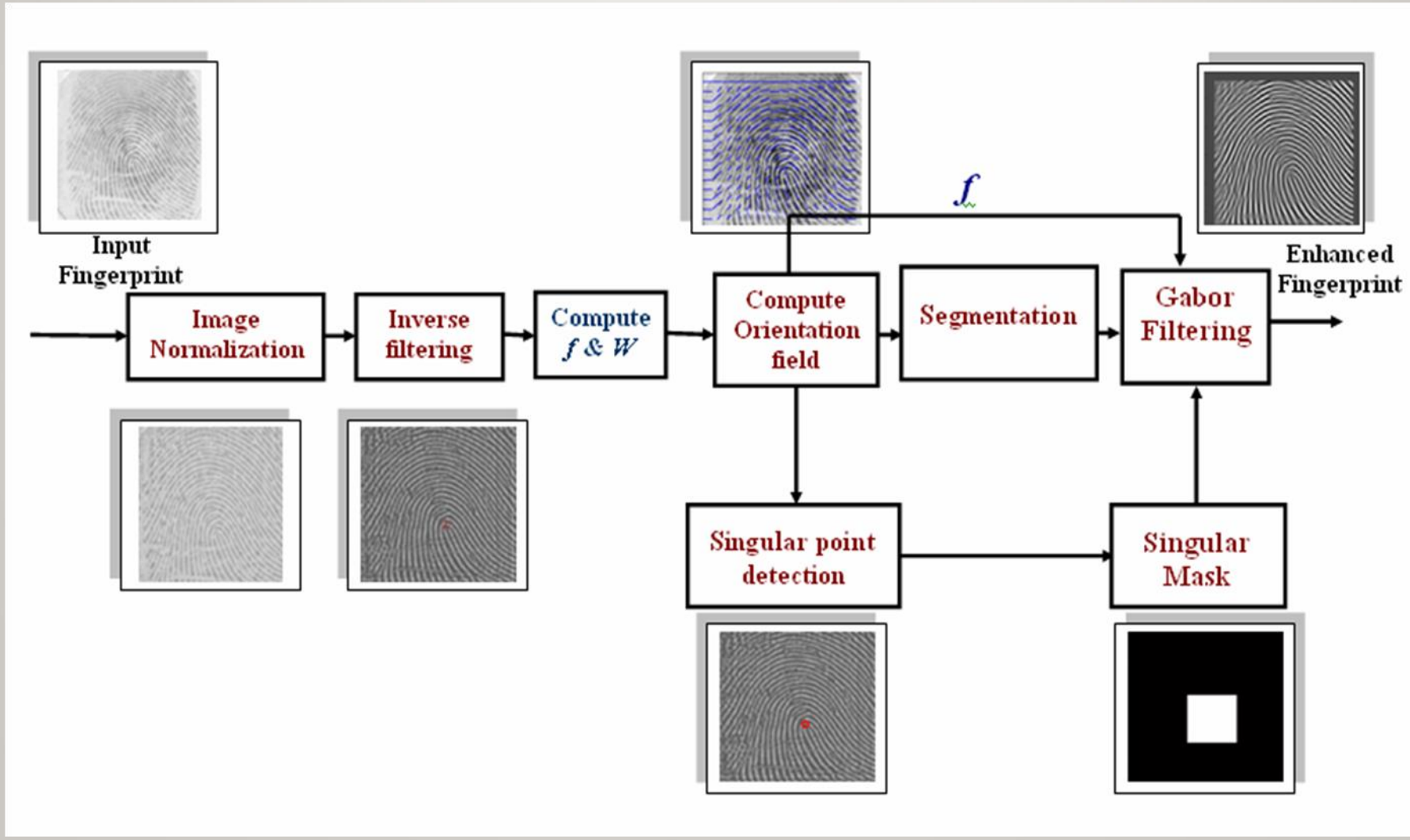
- 1. Segment
- 2. Erode to separate objects
- 3 Count Coins



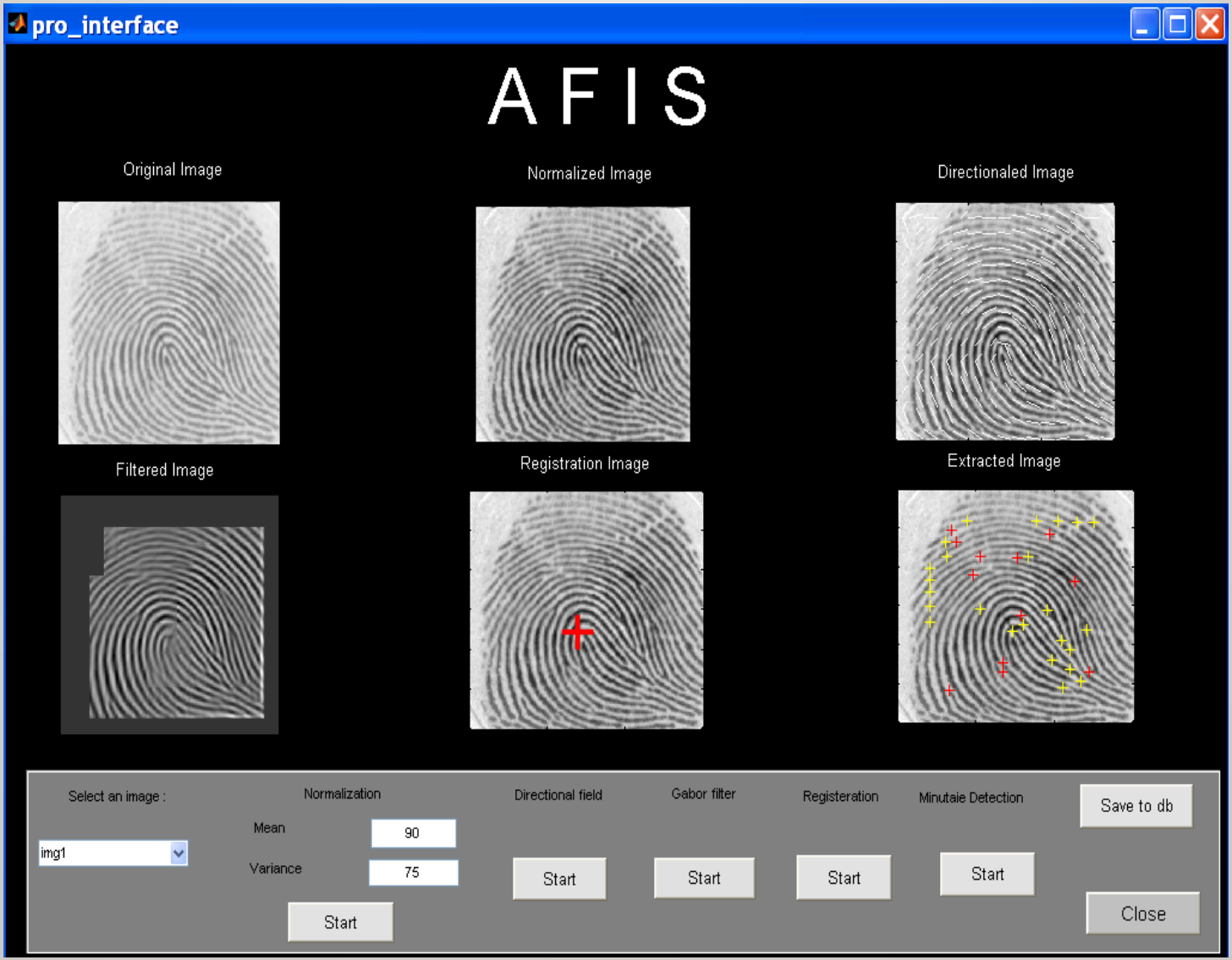
EXAMPLE OF DIP SYSTEM'



EXAMPLE OF DIP SYSTEM'



EXAMPLE OF DIP SYSTEM'



TYPES OF DIGITAL IMAGES
BINARY IMAGE

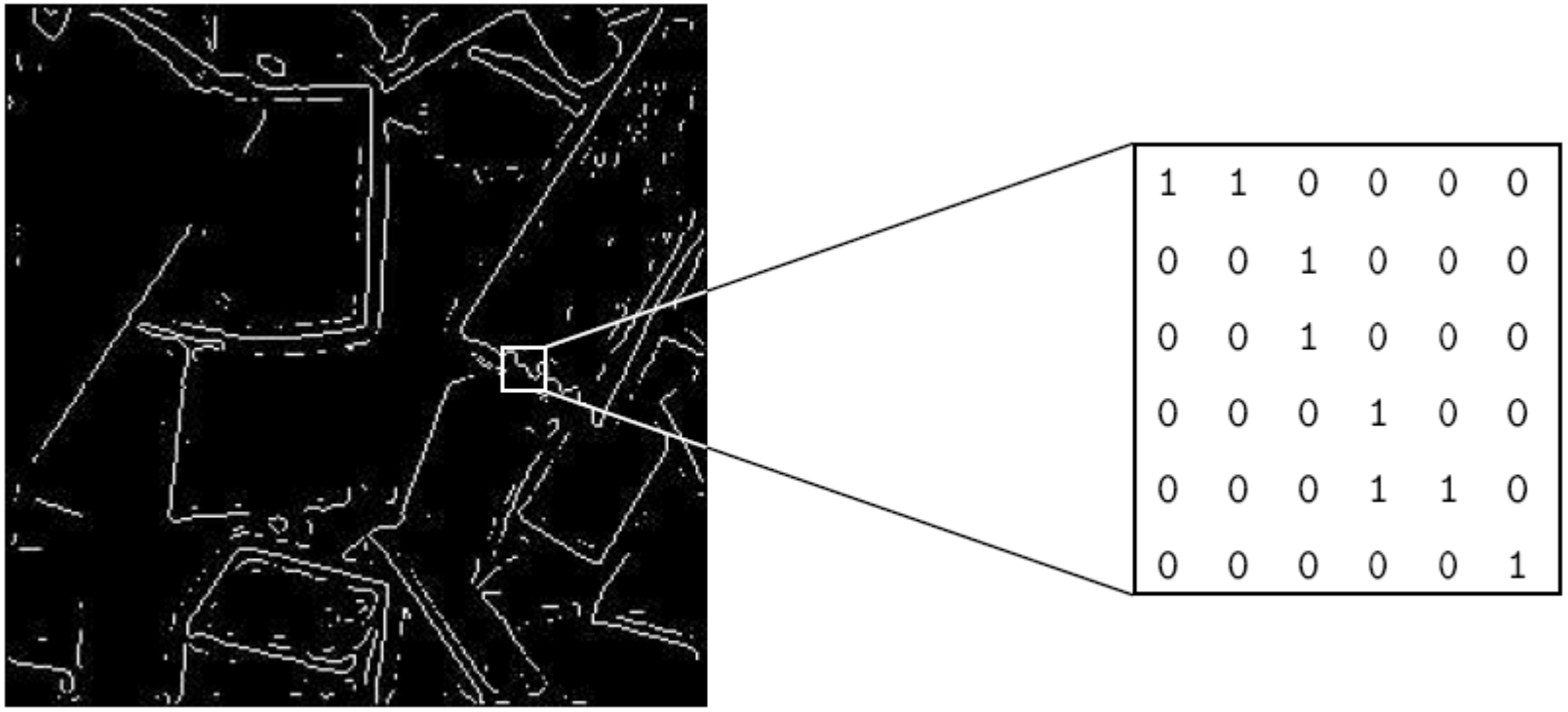


Figure 1.16: A binary image

1.8 TYPES OF DIGITAL IMAGES GRAYSCALE IMAGE



230	229	232	234	235	232	148
237	236	236	234	233	234	152
255	255	255	251	230	236	161
99	90	67	37	94	247	130
222	152	255	129	129	246	132
154	199	255	150	189	241	147
216	132	162	163	170	239	122

Figure 1.17: A greyscale image

1.8 TYPES OF DIGITAL IMAGES



49	55	56	57	52	53
58	60	60	58	55	57
58	58	54	53	55	56
83	78	72	69	68	69
88	91	91	84	83	82
69	76	83	78	76	75
61	69	73	78	76	76

Red

64	76	82	79	78	78
93	93	91	91	86	86
88	82	88	90	88	89
125	119	113	108	111	110
137	136	132	128	126	120
105	108	114	114	118	113
96	103	112	108	111	107

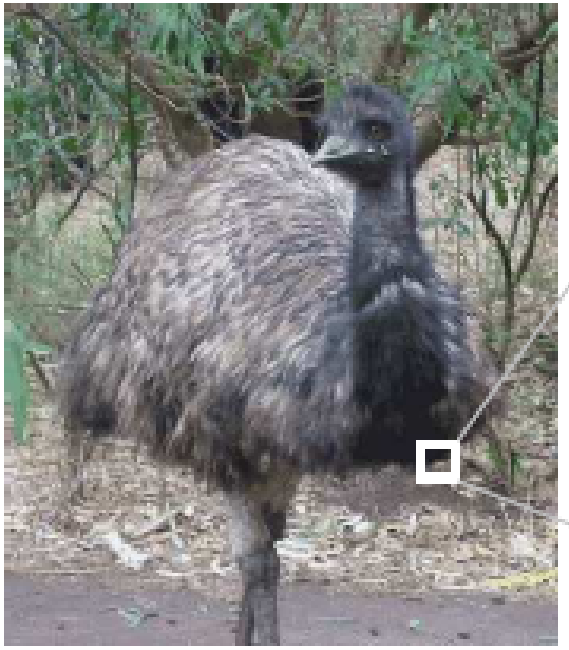
Green

66	80	77	80	87	77
81	93	96	99	86	85
83	83	91	94	92	88
135	128	126	112	107	106
141	129	129	117	115	101
95	99	109	108	112	109
84	93	107	101	105	102

Blue

Figure 1.18: A true colour image

1.8 TYPES OF DIGITAL IMAGES
INDEXED IMAGE



4	5	5	5	5	5
5	4	5	5	6	6
5	5	5	0	8	9
5	5	5	5	11	11
5	5	5	8	16	20
8	11	11	26	33	20
11	20	33	33	58	37

Indices

0.1211	0.1211	0.1416
0.1807	0.2549	0.1729
0.2197	0.3447	0.1807
0.1611	0.1768	0.1924
0.2432	0.2471	0.1924
0.2119	0.1963	0.2002
0.2627	0.2588	0.2549
0.2197	0.2432	0.2588
⋮	⋮	⋮

Colour map

Figure 1.19: An indexed colour image

CONVERTING FROM ONE IMAGE TYPE TO ANOTHER IN MATLAB

Function	Use	Format
ind2gray	Indexed to Greyscale	y=ind2gray(x,map);
gray2ind	Greyscale to indexed	[y,map]=gray2ind(x);
rgb2gray	RGB to greyscale	y=rgb2gray(x);
gray2rgb	Greyscale to RGB	y=gray2rgb(x);
rgb2ind	RGB to indexed	[y,map]=rgb2ind;
ind2rgb	Indexed to RGB	y=ind2rgb(x,map);

Table 1.2: Converting images in MATLAB

Resolution

Just like audio sampling, in image scanning we need to decide how many pixels we should record for each square or linear inch. The scanning rate in image processing is called **resolution**. If the resolution is sufficiently high, the human eye cannot recognize the discontinuity in reproduced images.

Color depth

The number of bits used to represent a pixel, its **color depth**, depends on how the pixel's color is handled by different encoding techniques. The perception of color is how our eyes respond to a beam of light. Our eyes have different types of photoreceptor cells: some respond to the three primary colors red, green and blue (often called **RGB**), while others merely respond to the intensity of light.

True-Color

One of the techniques used to encode a pixel is called True-Color, which uses 24 bits to encode a pixel.

Table 3.4 Some colors defined in True-Color

Color	*Red*	*Green*	*Blue*	*Color*	*Red*	*Green*	*Blue*
Black	0	0	0	Yellow	255	255	0
Red	255	0	0	Cyan	0	255	255
Green	0	255	0	Magenta	255	0	255
Blue	0	0	255	White	255	255	255

The **indexed color**—or **palette color**—scheme uses only a portion of these colors.

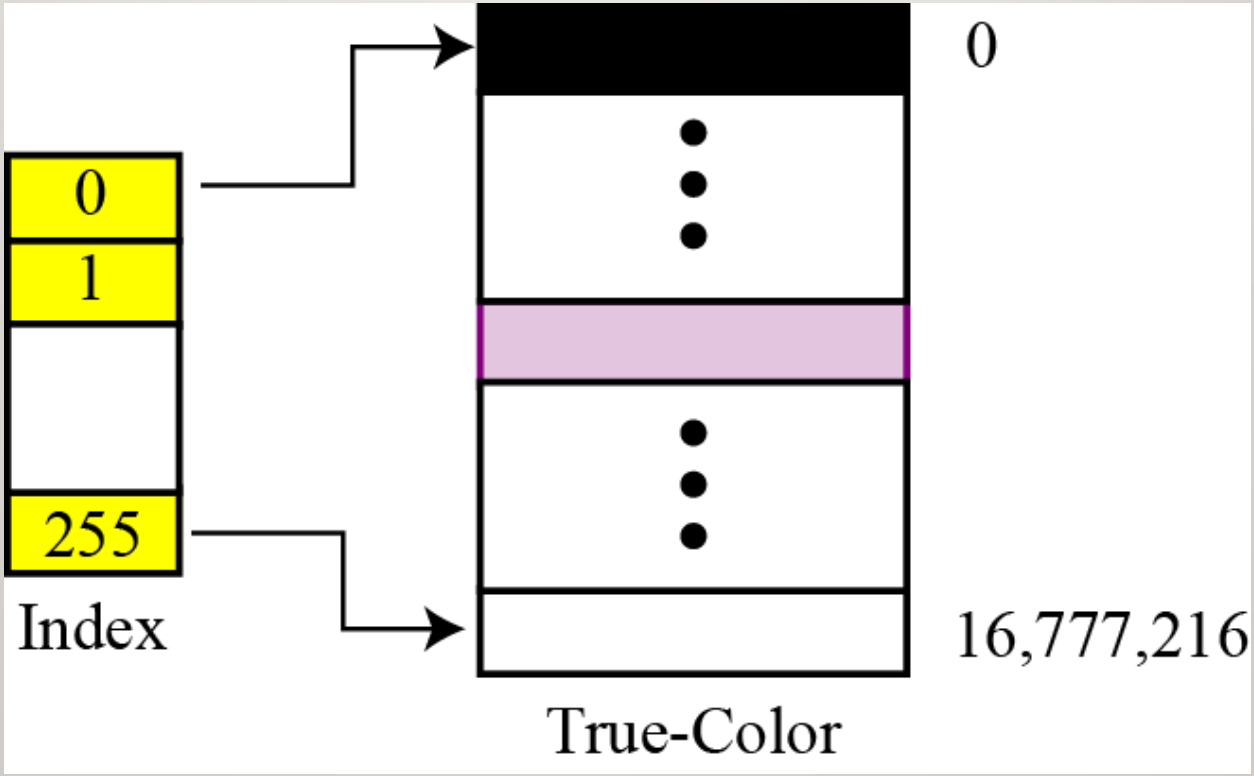


Figure 3.17 **Relationship of the indexed color to the True-Color**

For example, a high-quality digital camera uses almost three million pixels for a 3 × 5 inch photo. The following shows the number of bits that need to be stored using each scheme:

True-Color:	3,000,000	×	24	=	72,000,000
Indexed-Color:	3,000,000	×	8	=	24,000,000

Standards for image encoding

Several de facto standards for image encoding are in use. **JPEG** (**Joint Photographic Experts Group**) uses the True-Color scheme, but compresses the image to reduce the number of bits (see Chapter 15). **GIF** (**Graphic Interchange Format**), on the other hand, uses the indexed color scheme.

Lab Tasks:

- Read an image from files.
- Extracting image information.
- Converting from one image type to another.
- Note the effect of image conversion on image size and data type.

HOW TO CONVERT FROM GRAY SCALE TO BINARY IMAGE

$$Binary_iamge(i,j) = \begin{cases} 0 & if \;\; Gray_iamge(i,j) < Threshold \\ 1 & if \;\; Gray_iamge(i,j) >= Threshold \end{cases}$$

HOW TO CONVERT FROM RGB TO GRAYSCALE IMAGE

- Solution 1

$$Gray_image(i,j) = \frac{RGB_iamge_R(i,j) + RGB_iamge_G(i,j) + RGB_iamge_B(i,j)}{3}$$

- Solution 2

Choose one of R or G or B to represent the Gray image